

CSI3015 Software Project Management

Numerical Problems - Metrics

Number of Scenario Scripts (Use Cases)

Problem 1

A software team is developing an online library management system. The system must support operations for:

- Student
- Librarian
- Administrator

Each actor needs to perform exactly 4 unique use-case scenarios, and there are also 3 common system-wide scenarios used by all actors.

Find the total number of scenario scripts required.

Problem 2

An e-commerce platform initially contains 18 use-case scenarios. During refinement, 35% additional scenarios are added. Later, 4 obsolete scenarios are removed.

Calculate the final number of scenario scripts.

Problem 3

A mobile banking application includes:

- 12 essential use cases
- 8 optional use cases
- 25% more use cases added after customer feedback

Determine the updated number of scenario scripts.

Number of Key Classes (Class Diagram)

Problem 4

A system identifies the following domain elements:

- 9 entity classes
- 5 boundary classes
- 4 controller classes

During abstraction, 20% of the boundary classes are merged, and 1 new controller class is added.

How many key classes remain?

Problem 5

A class diagram for a hospital system has 28 initial classes. After applying design principles, 30% of the classes are removed due to redundancy, and 6 new essential classes are added.

Find the final number of key classes.

Problem 6

In a university ERP system, modules contribute key classes as follows:

- Admission: 7 classes
- Finance: 6 classes
- Examination: twice as many classes as Finance
- Hostel: 4 fewer classes than Admission

Compute the total number of key classes.

Number of Subsystems (Package Diagram)

Problem 7

A company develops a logistics platform with the following subsystems:

- Tracking
- Billing
- Customer Management
- Inventory

Each subsystem is split into 2 packages for maintainability. Later, Billing is further divided into 3 additional packages.

Determine the total number of packages (subsystems).

Problem 8

A distributed software project contains 15 subsystems initially. To support new business requirements, the number of subsystems increases by 40%. Later, 5 subsystems are merged into 2.

Find the final number of subsystems.

Problem 9

A smart-city application contains 6 major subsystems. Each subsystem contains an average of 4 minor subsystems. If 3 major subsystems double their minor subsystems due to expansion, **Calculate the new total number of subsystems.**

Numerical Problems for OO Metrics

Weighted Methods per Class (WMC)

Problem 1

A class Order Manager contains 8 methods. Their cyclomatic complexities are:

{3, 2, 5, 4, 3, 6, 2, 1}

Compute the **WMC**.

Problem 2

A class Customer Account has 10 methods.

- 6 simple methods with complexity **1** each
- 3 medium methods with complexity **4** each
- 1 complex method with complexity **10**

Find the **WMC**.

Problem 3

A software engineer reduces the WMC of class InvoiceService by refactoring.

Original WMC = 45

New method complexities total = 28

How much **WMC reduction** is achieved?

Depth of Inheritance Tree (DIT)

Problem 4

A class hierarchy is as follows:

Object $\rightarrow$ A $\rightarrow$ B $\rightarrow$ C $\rightarrow$ D

Find the **DIT for class D**.

Problem 5

A class PaymentGateway inherits from OnlineService, which inherits from BaseService. BaseService inherits from the system root class.

What is the **DIT for PaymentGateway**?

Problem 6

Given the inheritance tree depth for three modules:

- Module X: DIT = 3
- Module Y: 2 levels deeper than Module X
- Module Z: half the depth of Module Y (rounded down)

Find the **DIT values of X, Y, and Z**.

Number of Children (NOC)

Problem 7

A class Vehicle has the following subclasses: Car, Bike, Truck, Bus, Electric Car
Find the **NOC**.

Problem 8

A base class Employee has 12 direct subclasses. During restructuring, 5 classes are removed and 3 new subclasses are added.
Compute the **new NOC**.

Problem 9

A class Account originally had 7 children. After refactoring:

- 3 subclasses are merged into 1

- 2 new subclasses are introduced

Find NOC.

Coupling Between Object Classes (CBO)

1. A class OrderProcessor directly references:
 - 3 other classes via method calls
 - 2 classes via attribute declarations
 - 1 external library class

Calculate the CBO.

2. A class PaymentHandler has interactions with 12 classes. During refactoring, 5 couplings are removed and 3 new couplings are added. Find the final CBO.

3. Class UserService interacts with:
 - 6 classes through inheritance
 - 4 classes through method parameters
 - 2 classes through return types

If inherited couplings count as 50% of direct couplings, compute the effective CBO.

Response For a Class (RFC)

1. A class Invoice has:
 - a. 10 local methods
 - b. 4 methods in other classes that are invoked within these methods

Compute RFC.

2. A class Report Generator contains 15 methods. Each method calls 2 other class methods on average. **Calculate the RFC.**

3. Class AccountManager has:

- 18 methods
- Only 12 of them are actually invoked during runtime
- These 12 methods call 22 external methods

Find the RFC.

Lack of Cohesion in Methods (LCOM)

1. A class StudentRecord has 4 methods: M1, M2, M3, M4. Attributes A1 and A2 are used as follows:
 - M1 uses A1
 - M2 uses A2
 - M3 uses A1
 - M4 uses none

Total method pairs = 6, 4 pairs do **not** share any attribute. **Find LCOM.**

2. A class has 6 methods. Out of the 15 possible method pairs, 9 method pairs share at least one attribute. **Compute LCOM.**
3. A class Employee Data has: 5 methods
- Methods use attributes as follows:
 - $M1 \rightarrow \{A1, A2\}$
 - $M2 \rightarrow \{A2\}$
 - $M3 \rightarrow \{A3\}$
 - $M4 \rightarrow \{A1, A3\}$
 - $M5 \rightarrow \{A4\}$
 - Total method pairs = 10, 6 pairs **do not** share an attribute. **Find LCOM**